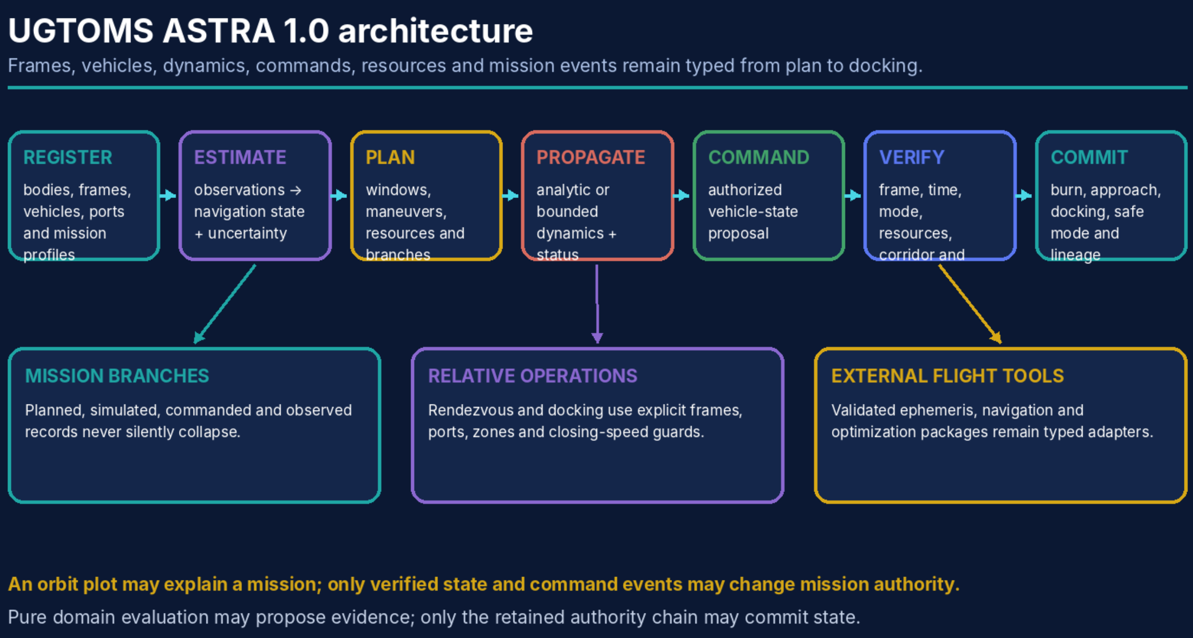
UNIVERSAL GEOMETRIC TOPOLOGICAL
OPEN MODULAR SUBSTRATE

UGTOMS ASTRA 1.0

Orbital Worlds, Spacecraft, Mission and Rendezvous Cartridge



`ugtoms.domain.astra@1.0.0`

Component	ASTRA
Canonical identity	ugtoms.domain.astra@1.0.0
Component type	Domain cartridge / orbital-mission state and event model
Version / date	1.0.0 / 30 August 2026
Document type	Open modular cartridge specification and implementation-ready profile
Status	Formal mission cartridge and deterministic synthetic fixture; not flight-certified guidance, navigation, control, collision avoidance or real mission authority

ATTRIBUTION AND EVIDENCE BOUNDARY

Prepared for Tom Klootwijk. Requester attribution is recorded as supplied and is not independently verified. This is a technical design artifact, not legal proof, scientific validation, regulatory approval or production safety certification.

Contents and Reading Rule

READING RULE

The UGTOMS source line supplies the reusable authority, typing, order, evidence and packing discipline. The specialist domain model in this cartridge is an engineering-derived extension. Where the source corpus is silent, this report says so instead of presenting general domain knowledge as source-derived.

Executive Decision and Release Scope

FORMAL DECISION

GO for ASTRA as the UGTOMS orbital-world and mission cartridge. Its signature move is not drawing pretty ellipses; it is preserving frame, epoch, units, uncertainty, vehicle identity, command authority, resources and event lineage through every estimate, propagation, maneuver, approach and docking transition.

ASTRA begins with a deterministic synthetic mission profile suitable for games, teaching, planning experiments and simulation orchestration. Real flight dynamics, navigation and operations are high-stakes specialist domains. The cartridge therefore treats validated ephemerides, estimators, optimizers and flight software as external authorities behind versioned adapters rather than pretending a short report replaces them.

- A numeric position or velocity is invalid without its reference frame, epoch, units and capability/uncertainty status.
- Planned, simulated, estimated, commanded and observed mission records are separate authority classes linked by explicit promotions.
- A maneuver or command is a proposal until current-state, resource, timing, authorization and conflict guards pass.

- Docking is a sequence of support zones and compatibility/kinematic guards, not a visual proximity event.

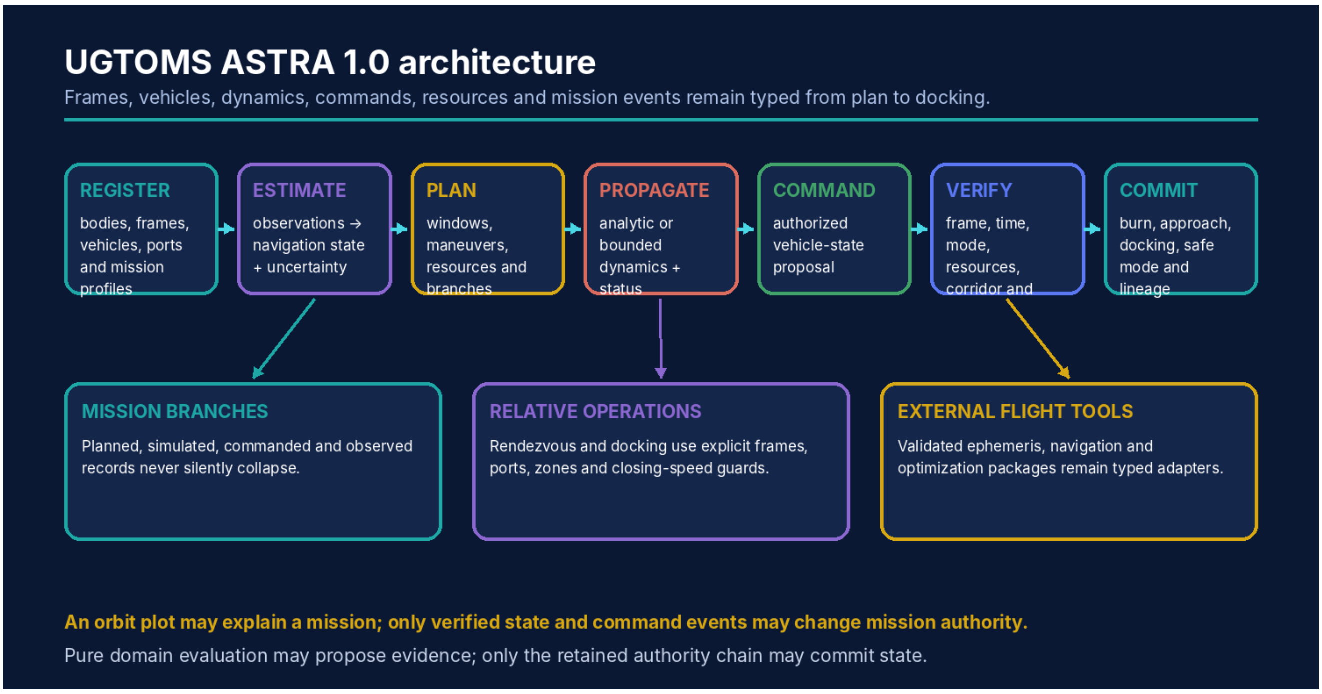


Figure 1. ASTRA keeps domain models upstream of the verified mission-event authority path.

Source Basis and Evidence Discipline

The cartridge inherits the following UGTOMS commitments:

- Typed definitions and operators live inside the substrate, with stable IDs, dependencies and content hashes [S1-S3].
- Pure evaluation is upstream of support, compatibility, guard, verified proposal, deterministic commit and lineage [S1, S2, S4].
- Coordinates, pictures, sounds, meshes and packed codes are not identity or authority by themselves [S4-S8].
- Uncertainty, exactness, capability and error margins are explicit and may only be strengthened by evidence [S1, S2, S8].
- Component-scoped identities prevent a specialist domain pack from masquerading as a universal replacement of every other release [S10].

ENGINEERING-DERIVED DOMAIN CONTENT

The celestial-body, frame, orbital-state, vehicle, maneuver, navigation, command, rendezvous, docking and spacecraft-system records in ASTRA are engineering-derived. The registered sources contribute frame/kinematic calculus, bounded dynamics, event order, stable identity, scenario branches, resources, proof/certificate discipline and runtime delivery [S1-S15]. No source report claims a ready aerospace standard or flight implementation.

ID	Registered source	Use in this cartridge	Pages
S1	UGTS__FOUNDATION__UNICODE_v5.pdf	Foundation 5.0 OperatorCells: literal, glyph, semantic, field, converse, packing and content-addressed faces remain typed.	pp. 1-21
S2	UGTS_KC_4_2_General_Operator_Order_Addendum(1).pdf	Typed OperatorSpec, dependency/phase/event/replay order, exactness, effects, capabilities, budgets and adapter boundaries.	pp. 2-32
S3	UGTS_KC_3_6_Tom_Klootwijk.pdf	Definitions as content-addressed substrate data, resolvable references, dependency closure and explicit operation order.	pp. 1-15
S4	Unified_Geometric_Topological_Substrate_GPU_Native_Addendum.pdf	Canonical support → compatibility → guard → verified event → transition → lineage authority and packed-state limits.	pp. 2-15
S5	UGTS_KC_2_0_Tom_Klootwijk_Expanded_Substrate.pdf	Patterns, fields, topology, SE(2)/SE(3), bounded dynamics, hybrid events and numerical/error contracts.	pp. 3-15
S6	UGTS_KC_Two_Hands_3_0_Interactive_Graphics_Runtime.pdf	Stable scene identity, proposals, deterministic commit, checkpoints, rollback, replay and divergence detection.	pp. 3-19
S7	UGTS_KC_3_9_KC_Elizabeth_Vector_Game_Runtime.pdf	Versioned projects, deterministic fixed-step runtime, snapshots/hashes and self-contained browser delivery.	pp. 2-16
S8	UGTS_KC_3_6_2_SCLP_Tom_Klootwijk.pdf	Time/phase/winding, finite topology maps, interval guards, finite grammar and exact packed-key boundaries.	pp. 1-15
S9	UGTS_Versioning_Charter_Phase_1_Foundation_Chronology(1).pdf	Component-scoped identities and explicit release graphs instead of one misleading global ladder.	pp. 1-6
S10	01_UGTOMS_REACT_1_0.pdf	Typed species/composition-like records, conservation checks, batch/evidence separation and external scientific adapters.	pp. 2-15
S11	04_UGTOMS_RULEFORGE_1_0.pdf	Reusable rule-world host: typed actions, visibility, chance, conflict resolution, terminality and deterministic replay.	pp. 2-18
S12	06_UGTOMS_PROOFWEAVE_1_0.pdf	Proof graphs, small kernel, scoped certificates, trust boundaries and proposal-only search.	pp. 2-18
S13	07_UGTOMS_CIVITAS_1_0.pdf	Layered world identity, route/support graphs, operational dependencies, observations, scenarios and verified city patches.	pp. 2-19
S14	08_UGTOMS_ARCANA_1_0.pdf	Typed fictional world laws, resource ledgers, executable symbols, support fields and rule-composition discipline.	pp. 2-20
S15	09_UGTOMS_CHRONOFORGE_1_0.pdf	Multiple clocks, immutable events, branch forks, loops, compensation, descendant invalidation and counterfactual replay.	pp. 2-20

Open Modular Cartridge ABI

The cartridge is a concrete instance of the shared domain object D_ASTRA:

```
D = (T, O, R, U, I, Q, E, G, Delta, L, Cert, K, Pi)
```

Field	Contract
T	Entity, value and state types
O	Typed operators and transformations
R	Relations, graphs, topology and converse pairs
U	Units, dimensions, coordinate and profile conventions
I	Invariants, conservation laws and constraints
Q	Dependency, precedence, phase and reduction order
E	Observations, evidence, uncertainty and provenance
G	Guards, thresholds and admission predicates
Delta	Proposed and committed state patches
L	Lineage, replay, history and migration
Cert	Exactness, residual, interval, scope and status certificates
K	Cold atlas, hot codebook and packed representation
Pi	Optional visual, audio, physical or device projection

A conforming cartridge contains a manifest, type registry, operator registry, relation atlas, unit/profile system, dependency DAG, guard library, transition definitions, claims ledger, examples and validation fixtures. External mature libraries may sit behind adapters when they are safer or more accurate than a fresh reimplementation.

General Operator Record and Order

```
OperatorSpec = (id, version, syntax, arity, domains, codomain, units, shape,
                laws, exactness, effects, capabilities, dependencies,
                order_contract, invariants, provenance, content_hash)
```

The record follows the General Operator and Order layer [S2]. A symbol or glyph is not allowed to smuggle in undeclared type, unit, order, inverse or authority semantics.

EXAMPLE RECORD

astra.execute-command consumes a signed/authorized command record, current vehicle state hash, clock/epoch, mission phase, resources and guard profile. It returns a rejected command or a finite vehicle/resource/state patch. A planner, user interface or ground model cannot directly mutate the vehicle or declare a burn complete.

Normative phase schedule

Phase	Meaning
1-3	parse, normalize and resolve content-addressed references
4-6	type, shape and unit validation; dependency closure
7-9	construct candidates; apply declared composition and reduction order
10-12	evaluate exact or bounded results; issue capability and residual certificate
13	support, compatibility and guard classification
14	verified proposal and deterministic conflict resolution
15	atomic commit, lineage, replay and optional projection

Formal Mission State and Authority Classes

```
q_AST = (P, F, B, O, V, M, S, C, E, H, L, X)

P mission/dynamics/navigation profiles   F frames, transforms, clocks and epochs
B celestial bodies and sites            O orbital/relative state records
V vehicles, ports and configurations    M plans, maneuvers, windows and branches
S propellant, power, thermal and modes  C commands, links and acknowledgments
E observations, estimates, uncertainty  H checkpoints + authoritative state hash
L burns, modes, docking and mission lineage X external tools and projections
```

Authority class	Examples	Promotion boundary
Planned	Desired trajectory, maneuver node, window, resource budget and alternatives.	Planning cannot write vehicle state.
Simulated	Propagated branch, predicted telemetry and hypothetical failures.	Remains model/profile-bound until explicitly selected or commanded.
Estimated	Navigation state from observations with covariance/status and provenance.	Estimate acceptance requires estimator and consistency guards.
Commanded	Authorized intent queued for a vehicle under current-state and timing conditions.	Still not executed or observed.
Committed/observed	Accepted burn/mode/dock event and linked telemetry/acknowledgment.	Only verified event commit changes authoritative mission state.

MISSION AUTHORITY SPLIT

plan != prediction != state estimate != command != executed event != observation. The records may correspond, but ASTRA requires a typed transition and lineage edge between them.

Frames, Bodies and Orbit Records

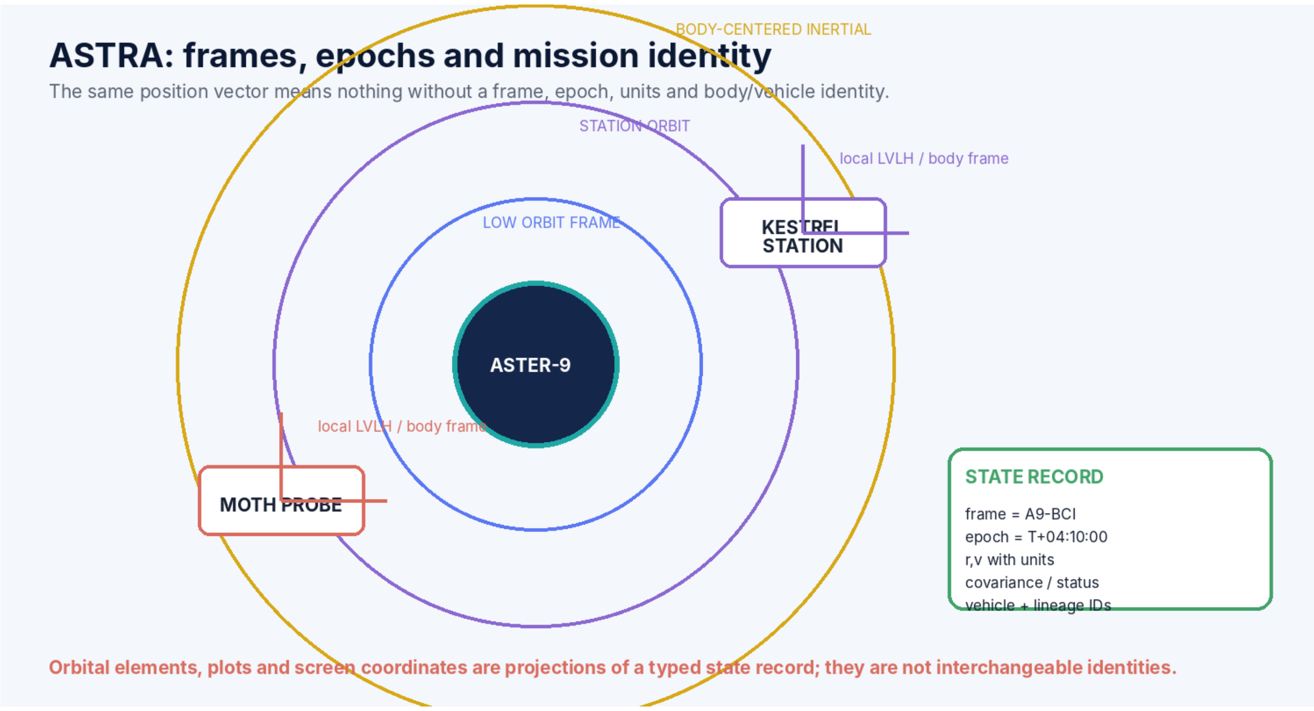


Figure 2. ASTRA treats frames and epochs as first-class identity context rather than annotations on a plot.

```
StateVector = (vehicle_id, frame_id, epoch, position[unit], velocity[unit/time],
               covariance_or_interval, capability, provenance, hash)
transform(state, target_frame, target_epoch) requires a declared transform provider
```

Record	Required fields	Failure mode prevented
Celestial body	Stable ID, gravity/shape/rotation/ephemeris capabilities, frame roots and validity.	A body name does not imply a specific gravity or ephemeris model.
Reference frame	Origin, axes, parent, transform provider, time validity and units.	Vectors from incompatible frames cannot be combined.
Epoch/time scale	Clock domain, scale, origin and conversion provider.	The same timestamp spelling cannot silently mean several time systems.
State vector	Vehicle, frame, epoch, position, velocity and uncertainty.	A plotted point never becomes an authoritative orbit by itself.
Orbital elements	Convention/profile-bound projection plus singularity/status.	Elements are not a universal identity or always numerically well-conditioned.

Propagation and Maneuver Operators

```
two-body profile:
    r_dot = v
    v_dot = -mu * r / |r|^3

impulsive proposal: v_plus = v_minus + delta_v(frame, epoch)
finite burn: integrate thrust, attitude, mass flow and forces with declared tolerances
```

Capability class	What may be claimed	What remains outside
Analytic two-body	Closed/bounded propagation under one central gravity parameter.	Perturbations, ephemeris uncertainty, maneuvers and navigation error.
Bounded numerical	Finite integration under named force model, step/error budget and terminal status.	Unmodeled forces and real-flight validation.
External high fidelity	Adapter result from named ephemeris/force/optimization software.	Authority still requires provenance, units, model version and mission policy.
Observed/estimated	Sensor-linked navigation state and uncertainty.	It is not exact physical truth and may diverge from prediction.

ORDER MATTERS

Frame transform, epoch conversion, burn application, propagation, observation assimilation and resource accounting are distinct phases. Swapping them may produce a different mission state even when every operator is individually legal.

Windows, Phasing and Rendezvous

Concept	Formal record	Guarded question
Window	Start/end interval, clock, target relation and admissible operation.	Is the proposal evaluated at a supported time and state?
Phasing	Relative orbital-state objective plus maneuver/coast sequence.	Does the branch enter the target region within resource/error limits?
Relative state	Target-local frame, relative position/velocity, epoch and uncertainty.	Are both vehicles transformed consistently?
Keep-out zone	Protected support volume and role/policy rules.	Is entry forbidden, authorized or unknown?
Hold point	Finite relative-state region requiring explicit release event.	Has navigation, command and corridor readiness passed?
Approach corridor	Geometry plus relative-speed, attitude, timing and port constraints.	Can the vehicle continue toward capture?

```
rendezvous_ok = frame_epoch_resolved AND target_support AND window_open
               AND relative_state_error <= margin AND resource_reserve
               AND branch/current_state_hash match
```

Spacecraft Systems, Resources and Modes

Subsystem	State and operators	Authority boundary
Propulsion	Propellant, thrust profile, valves/modes, burn cost and uncertainty.	A delta-v plan cannot spend mass until burn commit.
Power	Generation, storage, loads, margins and shedding priorities.	A predicted surplus is not a measured battery state.
Thermal	Temperatures, limits, attitudes, loads and thermal-model capability.	Simplified profiles cannot certify hardware safety.
Guidance/navigation/ control	Goals, estimates, commands, actuators and status.	Planner output, estimate and executed control remain separate.
Communications	Link windows, queues, delays, bandwidth and acknowledgments.	A transmitted command is not automatically received or executed.
Vehicle mode	Safe, cruise, maneuver, approach, docked or custom state machine.	Mode transitions are verified events with dependencies and reason codes.

```
resource_ok = propellant_after >= reserve_floor
          AND power_margin >= profile.minimum
          AND thermal_status in admitted_classes
          AND storage/communication capacity remains finite
```

Communications, Autonomy and Delayed Authority

Record	Meaning	Why it stays separate
Command proposal	Desired action, issuer, authorization, target state hash, execute window and expiry.	Intent is not execution.
Transmission event	Command bytes/message leave one endpoint under a link profile.	Transmission does not prove reception.
Receipt/acknowledgment	Vehicle accepts, rejects or queues the command.	Receipt does not prove completed actuation.
Autonomy rule	Onboard policy may generate proposals under bounded roles and modes.	Autonomy still uses the same verifier and resource guards.
Telemetry/observation	Measured report with timestamp, calibration, delay and provenance.	Delayed evidence must not overwrite newer state silently.
Reconciliation event	Ground and vehicle lineages compare and resolve known divergence.	Last-write-wins is not assumed.

DELAY IS PART OF STATE

Command time, receipt time, execution time, observation time and ground-ingest time may differ. ASTRA uses explicit clock domains and causal/dependency relations rather than one overloaded timestamp [S15].

Docking and Mission Commit Sequence

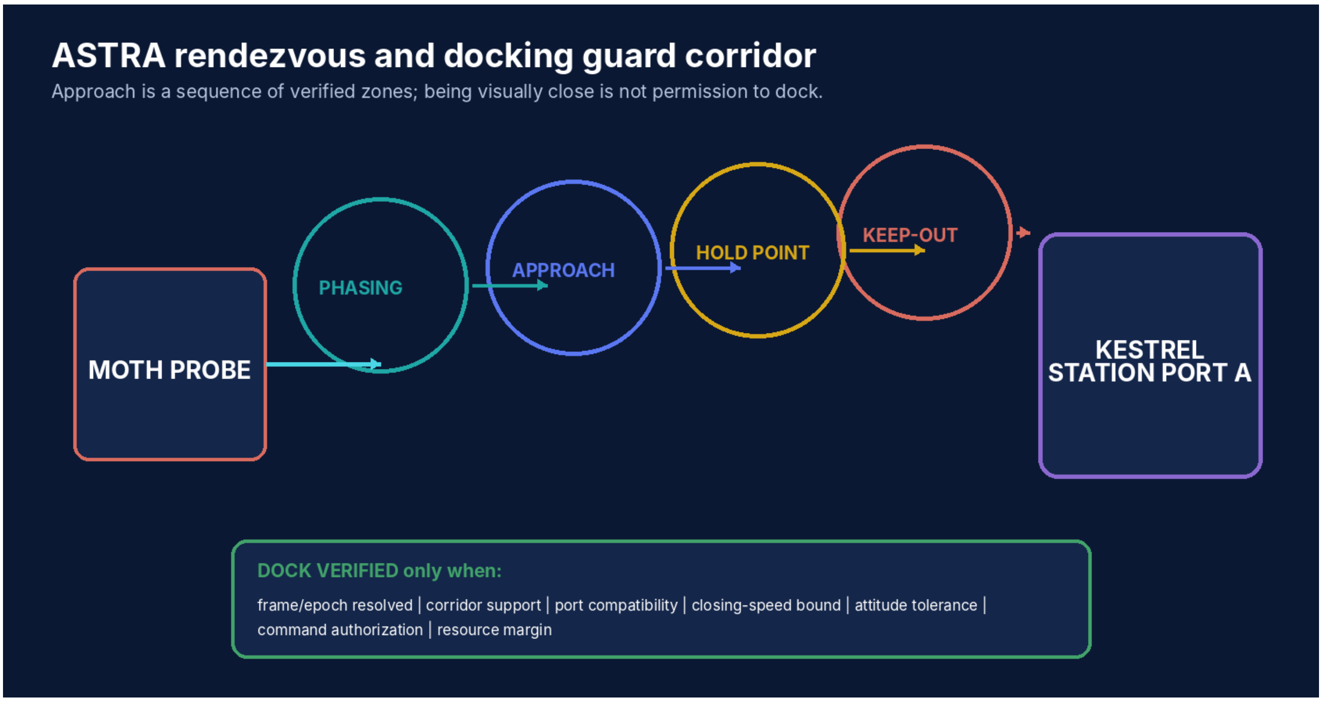


Figure 3. Docking authority is released through explicit zones and guards; visual proximity alone is insufficient.

```
dock_verified = port_ids_resolve AND port_compatibility
                AND relative_state in capture_envelope
                AND closing_speed <= limit AND attitude_error <= limit
                AND command_authorized AND resources/services ready
                AND no conflicting port occupancy event
```

Commit patch	Included state changes
Vehicle relation	Set docked-to relation and relative attachment frame.
Port state	Reserve/occupy compatible ports and record role/services.
Vehicle mode	Approach → docked or capture-latched under the profile.
Resource/services	Optional power/data/thermal/service links and their activation state.
Lineage	Command, estimate, guard results, contact/capture evidence and pre/post hashes.

PHYSICAL BOUNDARY

The synthetic docking guard demonstrates typed state and event order. It does not certify real contact dynamics, structural loads, proximity operations, sensor performance or crew safety.

Deterministic Kestrel Rendezvous Scenario

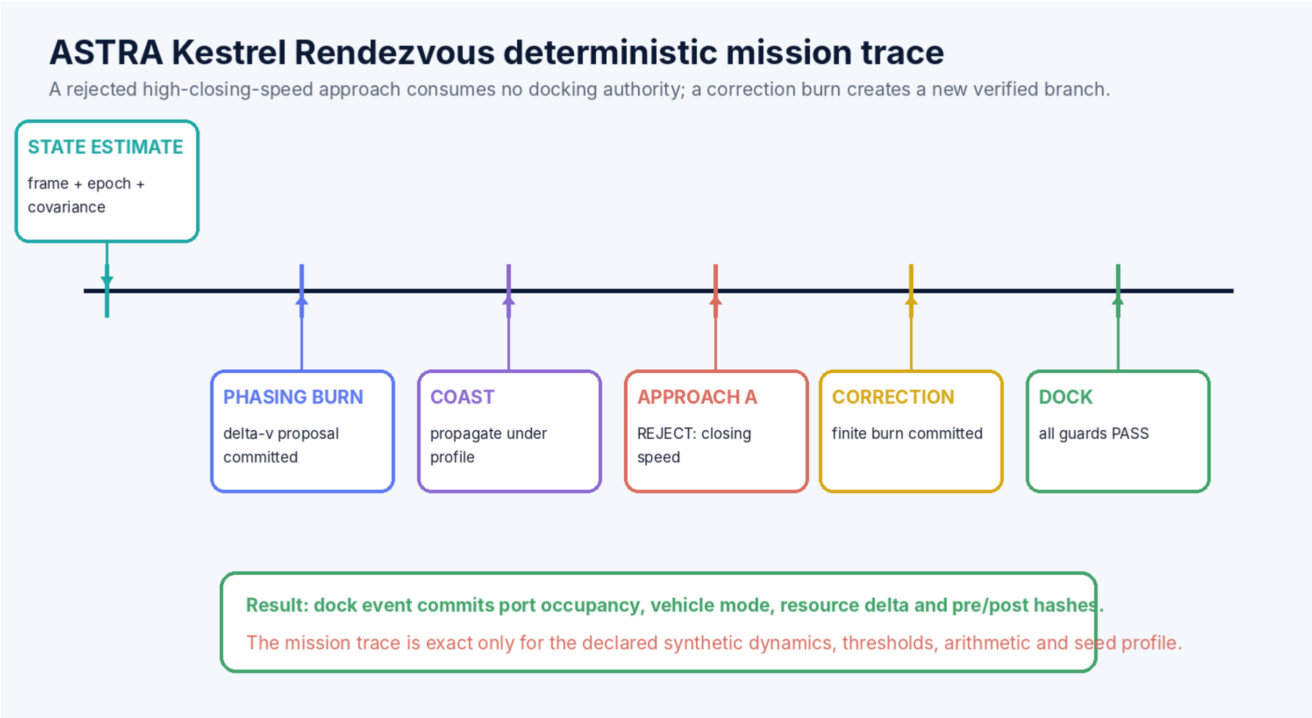


Figure 4. The first approach is rejected; a correction burn creates a new verified sequence ending in a dock commit.

```
fictional body: Aster-9; dynamics: normalized two-body profile
vehicles: Moth Probe -> Kestrel Station Port A
Approach A relative speed exceeds admitted capture corridor -> REJECT
Correction burn commits under current state/resource hash
Approach B passes corridor, attitude, port and authorization guards -> DOCK
```

Check	Fixture result
Frame/epoch integrity	All state vectors transform through registered A9 frames at explicit epochs.
Rejected command isolation	Approach A changes no port, vehicle mode or propellant authority.
Correction accounting	Burn event changes velocity and propellant together under one pre/post state patch.
Dock atomicity	Port occupancy, vehicle relation/mode and service state commit together.
Replay	Fixed arithmetic, profile, seed and event order reproduce identical hashes.

Storage, Queries, Adapters and Publication

Interface	Contract
Frame/body cold atlas	Content-addressed bodies, frames, transforms, vehicle/port definitions and dynamics capabilities.
Mission checkpoints + event deltas	State snapshots plus estimate, command, burn, mode, dock and observation lineages.
Mission query	Why a burn/dock was accepted, resource margin, active branch, pending commands, first divergence and unknowns.
Trajectory publication	Plots, ephemeris tables, timelines and browser views linked to exact source hashes and profiles.
CHRONOFORGE adapter	Counterfactual maneuvers, missed windows, faults and alternative mission branches.
VOLT/FORGE adapters	Typed vehicle systems, circuits, parts, interfaces and inspection state without duplicating specialist authority.
PROOFWEAVE adapter	Optional finite invariants for command order, resource bounds and synthetic rendezvous certificates.
External flight adapter	Validated ephemeris, navigation, optimization or mission-control tools with units/version/provenance.

PACKING BOUNDARY

Compact frame, body, maneuver and event indices reduce storage only when their global atlas and decode profile are present. A seed or hash cannot reconstruct unstored telemetry, disturbances or commands.

Reference Validation, Boundaries and Kill Criteria

REFERENCE SELF-CHECK RESULT

40/40 deterministic specification self-checks passed. These checks cover catalog uniqueness, required fields, dependency closure, demo arithmetic and declared reason codes. They are package evidence, not independent domain validation.

Checked area	Result
Catalog integrity	48 unique AST IDs with complete component-local contracts.
Frame/unit guards	Unresolved frame, epoch or unit combinations are rejected before dynamics.
Authority split	Plan, simulation, estimate, command and executed event remain distinct.
Rejected approach isolation	High-closing-speed approach changes no vehicle, port or resource authority.
Burn atomicity	Velocity and propellant changes commit together under one event.
Dock atomicity	Port occupancy, vehicle mode and attachment relation commit together.
Replay	Kestrel fixture reproduces identical event order and state hashes under the frozen profile.

Explicit non-claims

- No flight certification, mission readiness, real collision avoidance, guidance, navigation or control authority is claimed.
- No two-body or simplified numerical profile is claimed to model a real mission at operational fidelity.
- No state estimate is claimed to be exact physical truth; uncertainty and provenance remain mandatory.
- No content hash proves physical correctness, safety, fuel sufficiency or successful docking.
- No seed reconstructs unstored telemetry, disturbances, commands, failures or human decisions.

Kill criteria

1. Reject or narrow a use when a symbol, codebook slot or packed record cannot resolve to its exact global definition.
2. Reject when a capability or exactness label is stronger than the model, measurement or external source supports.
3. Reject when units, profiles, coordinate systems, operand order or dependency order are implicit.
4. Reject when a proposal bypasses support, compatibility, guards, deterministic commit or lineage.
5. Reject when an external adapter is required for safety or conformance but its version, calibration or provenance is absent.
6. Reject any performance, compression or safety claim without a named workload, target, error/equivalence contract and evidence.

Complete ASTRA 1.0 Mechanism Catalog

The catalog contains 48 cartridge-local mechanisms. These IDs are component-scoped and do not silently extend a global M-number ladder.

ID	Family	Mechanism	Core contract
AST001	Profile	Mission profile	Declares dynamics, frames, clocks, navigation, command, resource and authority semantics.
AST002	Identity	Celestial-body identity	Stable body definition with gravity/shape/ephemeris capability and provenance.
AST003	Identity	Vehicle identity	Stable spacecraft/probe/station record distinct from callsign, location and state estimate.
AST004	Identity	Mission identity	Content-addressed mission plan, rules, participants and release lineage.
AST005	Identity	Port identity	Docking/berthing interface with geometry, orientation, services and compatibility profile.
AST006	Frame	Reference frame	Origin, orientation, parent, transform provider, units and validity interval.
AST007	Frame	Epoch and time scale	Typed time coordinate; UTC-like display, mission elapsed and dynamics time remain distinct.
AST008	Frame	State vector	Position, velocity, frame, epoch, units and uncertainty/status.
AST009	Frame	Orbital elements projection	Profile-bound projection from state; singularities and convention are explicit.
AST010	Frame	Attitude state	Orientation, angular rate, frame, epoch and uncertainty.
AST011	Dynamics	Two-body propagator	Analytic/bounded state evolution under a declared central gravity profile.
AST012	Dynamics	Perturbation adapter	Optional higher-fidelity gravity, drag, radiation or multi-body model behind typed capability.
AST013	Dynamics	Impulsive maneuver	Instantaneous delta-v proposal under frame, mass, resource and event-order contracts.
AST014	Dynamics	Finite burn	Time-bounded thrust/mass-flow integration with terminal status and error budget.
AST015	Dynamics	Coast segment	Propagation interval with selected model, tolerances and checkpoints.
AST016	Dynamics	Relative-motion state	Target-relative position/velocity in a declared local frame and epoch.
AST017	Mission	Maneuver node	Planned burn with window, trigger, dependencies, alternatives and authority status.
AST018	Mission	Mission phase	Launch, cruise, approach, docked, surface, safe or custom state machine node.
AST019	Mission	Window relation	Finite interval in which a proposal may be evaluated; not a guarantee of feasibility.
AST020	Mission	Trajectory branch	Immutable planned/simulated/commanded branch with parent checkpoint.
AST021	Vehicle	Propellant ledger	Mass/resource quantities, reserves, uncertainties and consumption lineage.
AST022	Vehicle	Power ledger	Generation, storage, load, margin and operational modes.
AST023	Vehicle	Thermal state	Bounded temperatures/loads under a declared thermal adapter and safety profile.
AST024	Vehicle	Communication state	Link opportunities, delays, bandwidth, command queue and acknowledgment.

ID	Family	Mechanism	Core contract
AST025	Vehicle	Navigation estimate	Observation-derived state estimate and covariance/certificate; not physical truth by itself.
AST026	Operator	Propagate	State + time interval + model → bounded state/certificate.
AST027	Operator	Plan maneuver	Candidate delta-v/burn/window/resource patch without authority.
AST028	Operator	Execute command	Authorized command proposal evaluated against vehicle mode and current state hash.
AST029	Operator	Acquire navigation observation	Measurement proposal with sensor/calibration/time/provenance.
AST030	Operator	Estimate state	Observation set → navigation estimate under a named estimator adapter.
AST031	Relation	Orbits	Vehicle/body relation under a declared frame and dynamics profile.
AST032	Relation	Targets	Vehicle/mission to body, vehicle, site or port objective relation.
AST033	Relation	Communicates-with	Time-varying link support and compatibility relation.
AST034	Relation	Compatible-port	Mechanical, orientation, service, role and authorization compatibility.
AST035	Guard	Maneuver readiness	Frame, epoch, state-hash, mode, resource, timing and authorization pass.
AST036	Guard	Keep-out zone	Relative support and policy guard for protected approach regions.
AST037	Guard	Approach corridor	Position, relative velocity, attitude, timing and uncertainty remain within profile bounds.
AST038	Guard	Docking readiness	Port compatibility, capture envelope, services, command and conflict state pass.
AST039	Guard	Communication authorization	Origin, signature/profile, freshness and vehicle state permit command admission.
AST040	Invariant	Unit/frame consistency	No vector, time or quantity operation across unresolved units/frames/epochs.
AST041	Invariant	Resource nonnegativity	No command may commit unavailable propellant, power, storage or capacity.
AST042	Event	Burn commit	Atomic vehicle/resource/state transition with pre/post hashes and telemetry/command lineage.
AST043	Event	Dock commit	Atomic port occupancy, vehicle mode, relative-state and service relationship patch.
AST044	Event	Abort/safe transition	Reason-coded mode transition with retained mission context and irreversible markers.
AST045	Certificate	Trajectory segment certificate	Model, frame, epoch, interval, residual, status, resources and provenance.
AST046	Packing	Ephemeris/mission codebook	Local packed indices bound to global body/frame/operator atlas hashes.
AST047	Query	Mission and causality query	Windows, dependencies, resource margins, first divergence, command/telemetry lineage and unknowns.
AST048	Adapter	External astrodynamics/flight oracle	Validated ephemeris, navigation, optimization or mission software behind proposal-only typed ports.

Claims Ledger

Claim	Disposition	Reason
ASTRA can represent frames, epochs, state vectors, vehicles, maneuvers, resources and mission lineage.	ADMIT	The cartridge defines finite typed records and event contracts.
The same numeric vector is valid without a frame, epoch and units.	REJECT	Frame/time/unit identity is mandatory.
A two-body profile can be deterministic and analytically or numerically checked.	ADMIT	Only inside the declared central-body model and arithmetic/error contract.
A two-body result is a high-fidelity real mission prediction.	REJECT	Perturbations, ephemerides, navigation and operational constraints require validated adapters and evidence.
A rendered orbit proves spacecraft state.	REJECT	Plots are downstream projections and may hide frame/uncertainty/status.
A state estimate is identical to physical truth.	REJECT	It remains observation/model-derived and carries uncertainty and provenance.
A plan or simulation can directly command a vehicle.	REJECT	Authorization, current state, mode, guard and atomic command commit are required.
The docking guard can certify a synthetic fixture.	ADMIT	Within the declared relative-state, port, resource, arithmetic and threshold profile.
ASTRA is flight-certified guidance, navigation or control software.	REJECT	No hardware-in-the-loop, standards conformance, independent verification or safety certification is claimed.
Seeds can recreate unstored telemetry or external disturbances.	REJECT	External observations and novelty must be retained.
A content hash proves a trajectory is physically correct.	REJECT	It proves record identity/integrity only.
Mature mission tools may be integrated behind typed adapters.	ADMIT	Version, capability, units, model and provenance remain explicit.

High-Impact Application Profiles

Profile	What the cartridge enables
Orbital sandbox / space sim	Make frame-correct orbits, maneuvers, failures, resources and docking into transparent mechanics.
Mission-design puzzle game	Solve windows, phasing, communication delays and resource margins with replayable reasons.
Starship operations runtime	Compose VOLT, FORGE, CIVITAS-like habitat and CHRONOFORGE branch systems around one mission ledger.
Educational astrodynamics bridge	Move from normalized two-body fixtures to specialist tools without losing units, frames or authority boundaries.
Counterfactual mission lab	Fork failed approaches, missed windows and subsystem faults from immutable checkpoints.

Final Formal Definition

FORMAL DEFINITION

UGTOMS ASTRA 1.0 is a component-scoped orbital-world and mission cartridge in which celestial bodies, reference frames, epochs, state vectors, attitudes, vehicles, ports, dynamics capabilities, propagation segments, maneuvers, windows, mission phases, trajectory branches, navigation observations and estimates, commands, communication delays, propellant/power/thermal states, keep-out zones, approach corridors, docking compatibility, verified burns/mode transitions/dock events, checkpoints and mission lineage are typed substrate data; plans and simulations remain proposal-only; and only verified atomic mission events may change authoritative state.

Implementation sequence

1. Freeze Body, Frame, StateVector, Vehicle, Maneuver, MissionBranch, Command, DockEvent and SegmentCertificate schemas.
2. Implement the Kestrel fixture with frame/unit guards, rejected-approach isolation, burn accounting and dock atomicity.
3. Add a self-contained browser mission viewer where plots always link back to typed state and lineage.
4. Connect CHRONOFORGE, VOLT, FORGE and PROOFWEAVE through explicit adapter capabilities.
5. Admit real aerospace libraries only with validated version, units, frame/time conventions, model scope and proposal-only authority.

RELEASE CONCLUSION

This 1.0 document freezes a domain-facing contract. It is deliberately useful before a production engine exists: implementations can now disagree visibly through typed profiles, reason codes and certificates instead of hiding differences behind the same symbol.